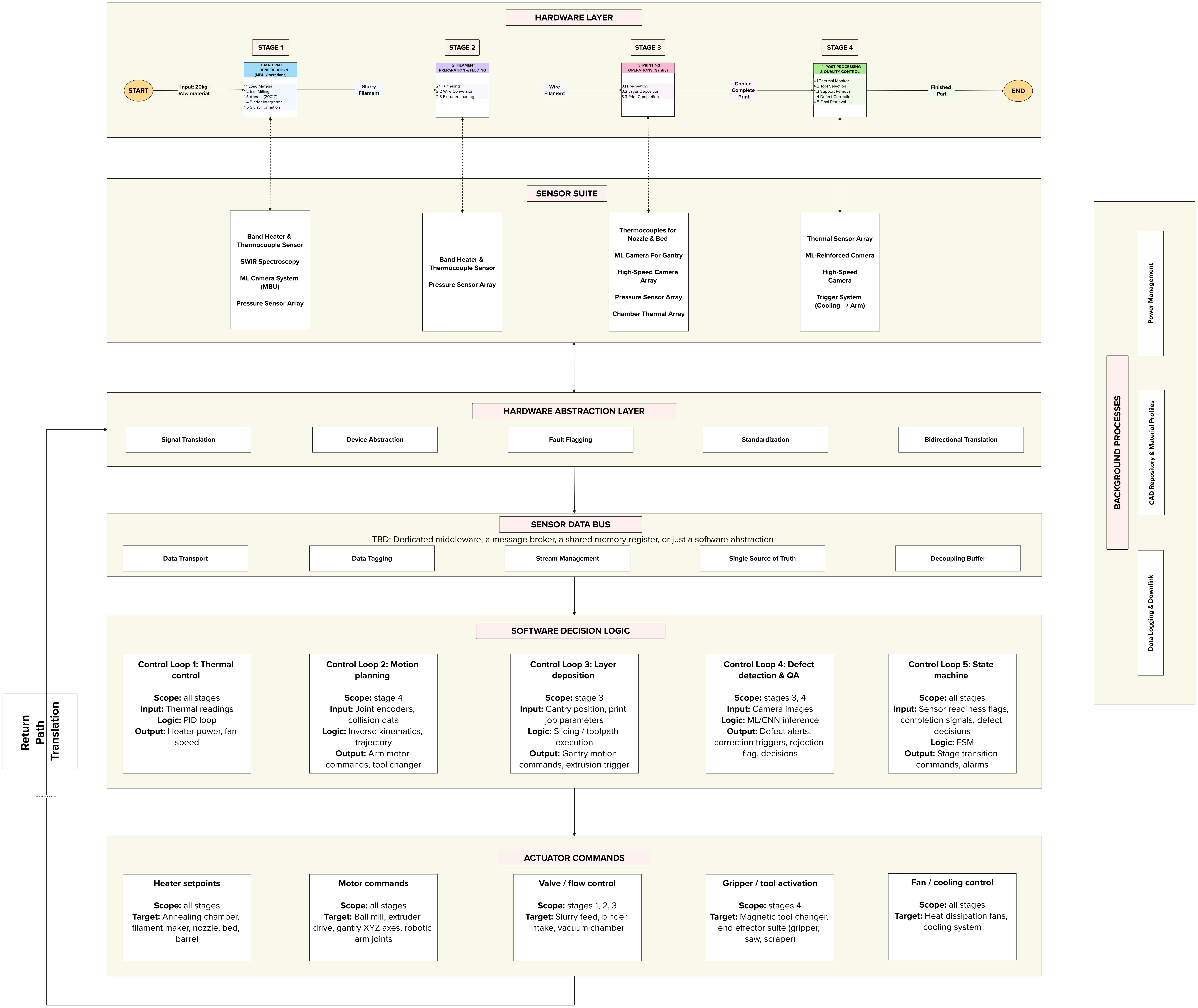
AERIS: System Architecture

- **Dashed line** (no arrow) = Physical connection
- **Dashed arrow** = Raw data flow
- **Solid arrow** = Processed flow



HARDWARE LAYER

START → Input: 20kg Raw material

STAGE 1

1. **MATERIAL BENEFICIATION (MBU Operations)**
1.1 Load Material
1.2 Ball Milling
1.3 Anneal (200°C)
1.4 Binder Integration
1.5 Slurry Formation

→ Slurry Filament

STAGE 2

2. **FILAMENT PREPARATION & FEEDING**
2.1 Funneling
2.2 Wire Conversion
2.3 Extruder Loading

→ Wire Filament

STAGE 3

3. **PRINTING OPERATIONS (Gantry)**
3.1 Pre-heating
3.2 Layer Deposition
3.3 Print Completion

→ Cooled Complete Print

STAGE 4

4. **POST-PROCESSING & QUALITY CONTROL**
4.1 Thermal Monitor
4.2 Tool Selection
4.3 Support Removal
4.4 Defect Correction
4.5 Final Retrieval

→ Finished Part → END

SENSOR SUITE

Band Heater & Thermocouple Sensor
SWIR Spectroscopy
ML Camera System (MBU)
Pressure Sensor Array

Band Heater & Thermocouple Sensor
Pressure Sensor Array

Thermocouples for Nozzle & Bed
ML Camera For Gantry
High-Speed Camera Array
Pressure Sensor Array
Chamber Thermal Array

Thermal Sensor Array
ML-Reinforced Camera
High-Speed Camera
Trigger System (Cooling → Arm)

HARDWARE ABSTRACTION LAYER

- **Signal Translation**
- **Device Abstraction**
- **Fault Flagging**
- **Standardization**
- **Bidirectional Translation**

SENSOR DATA BUS

TBD: Dedicated middleware, a message broker, a shared memory register, or just a software abstraction

- **Data Transport**
- **Data Tagging**
- **Stream Management**
- **Single Source of Truth**
- **Decoupling Buffer**

SOFTWARE DECISION LOGIC

Control Loop 1: Thermal control

Scope: all stages
Input: Thermal readings
Logic: PID loop
Output: Heater power, fan speed

Control Loop 2: Motion planning

Scope: stage 4
Input: Joint encoders, collision data
Logic: Inverse kinematics, trajectory
Output: Arm motor commands, tool changer

Control Loop 3: Layer deposition

Scope: stage 3
Input: Gantry position, print job parameters
Logic: Slicing / toolpath execution
Output: Gantry motion commands, extrusion trigger

Control Loop 4: Defect detection & QA

Scope: stages 3, 4
Input: Camera images
Logic: ML/CNN inference
Output: Defect alerts, correction triggers, rejection flag, decisions

Control Loop 5: State machine

Scope: all stages
Input: Sensor readiness flags, completion signals, defect decisions
Logic: FSM
Output: Stage transition commands, alarms

ACTUATOR COMMANDS

Heater setpoints

Scope: all stages
Target: Annealing chamber, filament maker, nozzle, bed, barrel

Motor commands

Scope: all stages
Target: Ball mill, extruder drive, gantry XYZ axes, robotic arm joints

Valve / flow control

Scope: stages 1, 2, 3
Target: Slurry feed, binder intake, vacuum chamber

Gripper / tool activation

Scope: stages 4
Target: Magnetic tool changer, end effector suite (gripper, saw, scraper)

Fan / cooling control

Scope: all stages
Target: Heat dissipation fans, cooling system

Return Path Translation

Return Path Translation

BACKGROUND PROCESSES

- **Power Management**
- **CAD Repository & Material Profiles**
- **Data Logging & Downlink**